

# Arunabha Majumder

## Mechanical Engineer — Wearable Devices, Exoskeletons, Sensor Integration & Rapid Prototyping

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### SUMMARY

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Mechanical engineer for body-worn devices. Designed and built two exoskeletons, from mechanical architecture and sensor integration to rapid 3D-printed iterations and testing with human participants. Force-sensing method validated to below 10% error; wearable design that cut energy use by more than 25% (Best Paper Award, 2026).

### KEY RESULTS

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- **>25%** energy reduction from the wearable's actuation architecture
- **>15%** lower wearer muscle effort, tested with human participants
- **<10%** RMS error on force sensing, validated against a force sensor

### EXPERIENCE

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#### Research Engineer – Wearable Robotics & Mechanical Architecture (PhD Project: VIE XO) | 2023 – Present

*Aalborg University, Aalborg, Denmark*

- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by owning the mechanical architecture of a wearable shoulder exoskeleton: choosing a hybrid actuation concept that pairs a parallel spring with a 6 Nm motor, balancing performance, component packaging and weight on the body.
- Took the wearable from early prototype to tested system by designing it in SolidWorks and iterating rapidly through 3D-printed prototypes based on hands-on testing.
- Integrated sensors and embedded components (motor, encoder, force sensors, DAQ and embedded control, with cabling) into one working device, working across mechanical, electronics and control.
- Achieved below 10% RMS error in force sensing, validated against a force sensor, by developing a force-estimation method on a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian; applied it to measure human arm impedance in the gait lab.
- Reduced wearer muscle effort by more than 15%, measured with EMG, by implementing an Assist-as-Needed controller tested with human participants; conducted data collection with 12 participants across 3 modes and 2 load conditions.
- Mentored and supervised Bachelor's (B.Tech) students (2023 – 2025).

#### MTech Researcher – Soft Pneumatic Actuators (Master's Thesis) | 2020 – 2022

*CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Reduced position-tracking error to 0.3–0.78% with a neural-network-based controller for soft pneumatic artificial muscles, which I fabricated from raw materials along with their test rig (valves, pressure sensors, DAQ).

#### Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton (Wearable Device) | 2018 – 2019

*Siddaganga Institute of Technology, Tumakuru, India*

- Won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant by leading CAD design, machining, hands-on assembly and user testing of a wearable rehabilitation device.

### SKILLS

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- **Wearables:** Exoskeletons, body-worn mechanisms, physical human-robot interaction, testing with participants

- **Sensor & electronics integration:** Force sensors, encoders, motors, DAQ, embedded control, cabling, EMG, Qualisys motion capture
- **CAD & prototyping:** SolidWorks (CSWA), 3D printing, rapid iteration; machining and assembly (Bachelor's)
- **Analysis & control:** Kinematics, force estimation, Assist-as-Needed control, MATLAB/Simulink, Python
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

## EDUCATION

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**Ph.D. Candidate, Mechatronics & Robotics** | 2023 – Present

*Aalborg University, Department of Materials and Production, Aalborg, Denmark*

**M.Tech, Mechatronics** | 2020 – 2022

*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

**B.Tech, Mechanical Engineering** | 2015 – 2019

*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India*

## AWARDS & PUBLICATIONS

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- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

## REFERENCES

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Available upon request.